

LEARNING PLAN

Unit Title: SECURITY SYSTEM INSTALLATION	Unit code: ENG/CU/EI/CR/02/5/MA
Name of trainer: MARGARET	
Institution: ABERDARE TECHNICAL COLLEGE	Level:
Date of preparation: 28/8/2026	Date of revision:
Number of learners:	Class: MODULE 2
Skill or job task: 1. Apply health and safety measures. 2. Install security systems. 3. Test security system installation. 4. Maintain security system installation.	
Bench marks or performance criteria: 1. Apply Health and Safety Measures 1.1 Relevant clauses in appropriate Acts are identified. 1.2 Occupational Safety and Health Act (OSHA) requirements are observed. 1.3 Work Injury Benefits Act (WIBA) provisions are observed. 1.4 Safety regulations and procedures are applied. 1.5 Work site is prepared according to installation requirements. 1.6 Security system installation tools are identified and used appropriately. 1.7 Fixing tools are identified and used. 1.8 Measuring tools are identified and used.	

2. Install Security Systems

- 2.1 Types of cables used in security systems are identified.
- 2.2 Factors considered during security system cable laying are explained.
- 2.3 Cable segregation requirements are applied.
- 2.4 Importance of cable segregation is explained.
- 2.5 Security systems are installed according to specifications.
- 2.6 CCTV system materials and components are identified.
- 2.7 CCTV cables are identified.
- 2.14 Size and class are considered.
- 2.15 CCTV wiring diagrams and schematic diagrams are interpreted.
- 2.16 CCTV systems are wired according to specifications.
- 2.17 Alarm systems are identified.
- 2.18 Fire alarm systems are installed.
- 2.19 Burglar alarm systems are installed.

3. Test Security System Installation

- 3.1 Security system installation is visually inspected.
- 3.2 Types of security system tests are identified.
- 3.3 Insulation tests are performed.

3.4 Short circuit tests are performed.

3.5 Continuity tests are performed.

3.6 Arming and disarming tests are performed.

3.7 Physical inspection of the security system is conducted.

3.8 Test results are documented

4. Maintain Security System Installation

4.1 Maintenance schedules are prepared.

4.2 System maintenance checklists are prepared.

4.3 Appropriate maintenance tools and equipment are selected.

4.4 Security systems are inspected and tested.

4.5 Faults are diagnosed.

4.6 Identified faults are rectified.

4.7 Maintenance reports are documented.

4.8 Security system maintenance is performed according to applicable regulations and procedures.

Wee k	Sessio n No.	Session Title	Specific Learning outcomes	Key Learning Points	Trainee Activities	Learning Resources	Learning checks/ assessment	Reflectio n's & Date
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1	1-2	Apply Health and Safety Measures	By the end of the session, the trainee should be able to: 1. Explain relevant health and safety requirements. 2. Identify OSHA and WIBA provisions applicable to security system installation. 3. Apply safety regulations and procedures at the worksite.	• OSHA requirements • WIBA requirements • Safety regulations • Safety procedures • Hazards associated with electrical and security system installation • PPE requirements	• Discuss relevant safety legislation. • Identify hazards in a simulated worksite. • Select and wear appropriate PPE. • Demonstrate safe working procedures.	PPE, safety charts, first aid kit, fire extinguisher, OSHA/WIBA reference materials, whiteboard		
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	3–4	Worksite Preparation and Security Installation Tools	1. Prepare a security system installation worksite. 2. Identify fixing, measuring, termination and cutting tools. 3. Demonstrate proper use and care of tools.	• Worksite preparation • Fixing tools • Measuring tools • Crimping tools • Cutting tools • Tool care and maintenance	• Prepare a simulated installation site. • Identify tools and state their functions. • Demonstrate correct handling and storage of tools.	Screwdrivers, drills, pliers, crimping tools, cable cutters, tape measures, tool manuals		
2	1–2	Electrical Measuring Instruments and Marking Tools	1. Identify electrical measuring instruments. 2. Use a multimeter, earth resistance meter and cable tester. 3. Apply correct marking procedures.	• Multimeter • Earth resistance meter • Cable tester • Scribes • Pliers • Importance of accurate marking	• Measure electrical quantities using instruments. • Test sample cables. • Mark installation positions according to a layout.	• Multimeters, earth resistance meters, cable testers, scribes, measuring tapes, marking materials	Practical demonstration; written test; oral questioning	

	3–4	FORMATIV E ASSESSME NT 1	1. Demonstrate application of health and safety measures. 2. Select and use tools correctly. 3. Prepare a worksite safely.	• Safety procedures • PPE • Tool identification • Measuring instruments • Worksite preparation	• Complete practical safety and tool identification tasks. • Answer oral questions. • Submit portfolio evidence.	Assessment tools, PPE, installation tools, rubrics, question papers	Knowledge: written/oral test; Skills: practical assessment; Attitude: observation	
3	1–2	Security System Cables and Cable Laying	1. Identify types of security system cables. 2. Explain factors considered during cable laying. 3. Apply cable segregation requirements.	• Types of security cables • Cable selection • Cable routing • Cable segregation • Importance of segregation	• Identify different cable types. • Plan cable routes. • Demonstrate proper cable segregation.	Security cables, cable samples, trunking, conduits, installation drawings	Practical; project; oral questioning; written test	

	3–4	Introduction to CCTV Systems	1. Identify CCTV system components. 2. Identify CCTV cables and containment systems. 3. Explain data and power backup requirements.	• CCTV cameras • DVR/NVR • Monitors • Network equipment • Cables • Conduits and trunking • Data backup • Power backup	• Identify CCTV components. • Match components to their functions. • Develop a simple CCTV installation layout.	CCTV cameras, DVR/NVR, monitors, cables, UPS, installation manuals	Practical demonstration; oral test; written test	
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4	1-2	CCTV Tools, Equipment and Materials Specification	1. Select CCTV installation tools and materials. 2. Interpret tolerance, range, make/model, size and class specifications .	• Tool selection • Material specifications • Tolerance/range • Make/model • Size • Class	• Compare tools and equipment specifications . • Select suitable materials for a given installation task.	Manufacturer manuals, CCTV catalogues, tools, sample components	Practical assessment; project; oral questioning	
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	3–4	CCTV Wiring Diagrams and Installation	1. Interpret CCTV schematic diagrams. 2. Interpret wiring diagrams. 3. Install and wire a CCTV system.	• CCTV schematic diagrams • Wiring diagrams • Cable routing • Camera connections • DVR/NVR connections	• Read installation drawings. • Route cables. • Terminate cables. • Connect CCTV components.	CCTV training board, cameras, DVR/NVR, cables, connectors, tools	Practical assessment; project; observation	
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5	1-2	CCTV Configuration and Testing	1. Configure CCTV equipment. 2. Test camera operation. 3. Configure recording and monitoring functions.	• Camera configuration • Network settings • Recording • Storage • Remote monitoring • Functional testing	• Configure CCTV equipment. • Test video output. • Set recording parameters. • Demonstrate system operation.	CCTV system, computer, network equipment, configuration manuals	Practical test; oral questioning; portfolio evidence	
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	3–4	FORMATIV E ASSESSME NT 2	1. Demonstrate CCTV installation skills. 2. Interpret wiring diagrams. 3. Configure and test a CCTV system.	• CCTV components • Wiring • Termination • Configuration • Testing	• Complete a practical CCTV installation task. • Answer oral and written questions.	CCTV equipment, tools, assessment rubrics, question papers	Knowledge: written/oral test; Skills: practical assessment; Attitude: observation	
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6	1-2	Introduction to Alarm Systems	1. Identify types of alarm systems. 2. Explain the operation of fire and burglar alarm systems. 3. Identify alarm system components.	• Fire alarms • Burglar alarms • Sensors • Control panels • Sounders • Detectors • Backup power	• Identify alarm components. • Classify alarm systems. • Demonstrate basic alarm system operation.	Fire alarm panel, burglar alarm panel, detectors, sensors, sounders	Practical; oral questioning; written test	
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	3–4	Alarm System Materials and Wiring	1. Select alarm system materials. 2. Interpret alarm system schematic and wiring diagrams. 3. Wire alarm system components.	• Alarm cables • Conduits and trunking • Alarm components • Backup power • Wiring diagrams	• Interpret diagrams. • Install containment systems. • Route and terminate cables. • Connect alarm components.	Alarm training boards, control panels, cables, conduits, tools	Practical assessment; project; observation	
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7	1-2	Fire Alarm System Installation	1. Install fire alarm components. 2. Connect detectors, call points and sounders. 3. Test basic fire alarm operation.	• Fire alarm control panel • Smoke detectors • Heat detectors • Manual call points • Sounders • Zoning	• Install fire alarm components. • Wire a simple fire alarm circuit. • Conduct functional tests.	Fire alarm panel, detectors, call points, sounders, cables	Practical; project; oral questioning	
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	3–4	Burglar Alarm System Installation and Configuration	1. Install burglar alarm components. 2. Configure alarm zones. 3. Test arming and disarming functions.	• PIR sensors • Magnetic contacts • Control panels • Keypads • Zones • Arming/disarming	• Install sensors and contacts. • Wire the control panel. • Configure zones. • Test arming and disarming.	• Burglar alarm panel, PIR sensors, contacts, keypad, backup battery	Practical assessment; oral questioning; portfolio	
8	1–2	Maintenance Planning and Inspection	1. Prepare a maintenance schedule. 2. Develop a maintenance checklist. 3. Inspect security systems.	• Preventive maintenance • Maintenance schedules • Checklists • Inspection procedures	• Prepare a maintenance schedule. • Develop an inspection checklist. • Conduct a routine inspection.	Maintenance manuals, sample schedules, checklists, security systems	Project; practical; written test	
10	1–4	SUMMATIVE CDACC ASSESSMENT	The trainee demonstrates competence in: 1. Applying health and safety measures. 2. Installing security systems. 3. Testing	Integrated assessment of all learning outcomes.	• Perform assigned practical task. • Complete written/oral assessment. • Submit portfolio evidence where applicable.	Assessment tools, CCTV systems, alarm systems, electric fence systems, tools, PPE, assessment rubrics	Practical Assessment ; Project; Written Test; Oral Questioning; Portfolio of Evidence; Third Party Report where applicable	

			security system installatio ns. 4. Maintaini ng security system installatio ns.					
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Prepared by: MD.Margaret
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